

Exercises in Marine Ecological Genetics

11. Metabarcoding / eDNA II

- Alpha diversity
- Beta diversity
- Community composition

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<https://github.com/mhelmkampf/meg26>

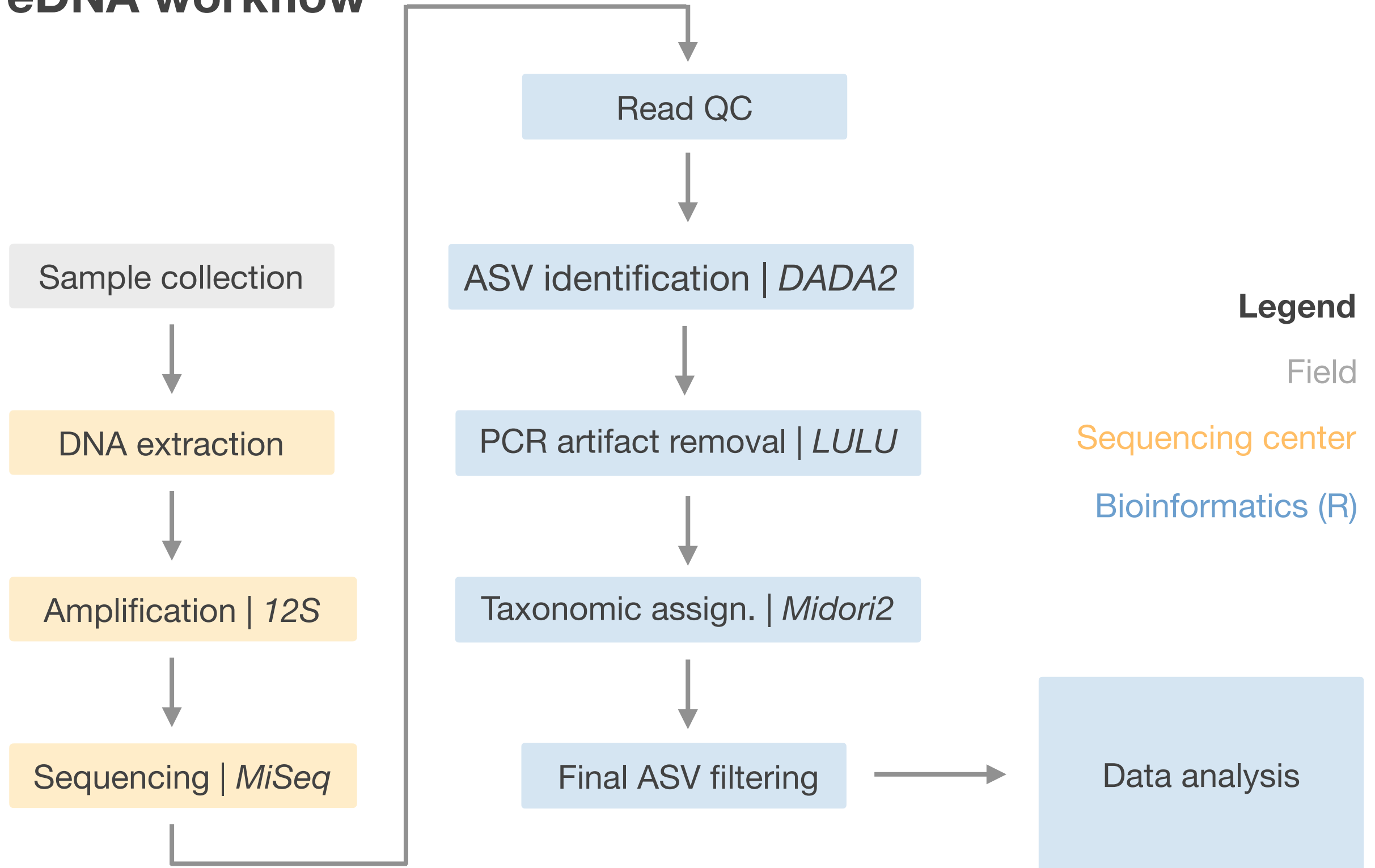


Today's objectives

By the end of this session, you will be able to:

- Calculate and interpret **alpha diversity** metrics (e.g., species richness and Shannon diversity) from eDNA metabarcoding data to compare within-sample biodiversity across sites or conditions
- Quantify and interpret **beta diversity** among eDNA samples using dissimilarity metrics and ordination methods to evaluate differences in community composition
- Construct and interpret stacked bar plots of taxonomic composition to identify dominant taxa and compare patterns in community structure across variables

eDNA workflow



Monitoring seagrass restoration

- Restoration began **July 2025**
- Transplantation of single shoots and **plugs** tested at different densities
- eDNA sampling after **0, 3, 6, 12 and 18 months**



Dataset

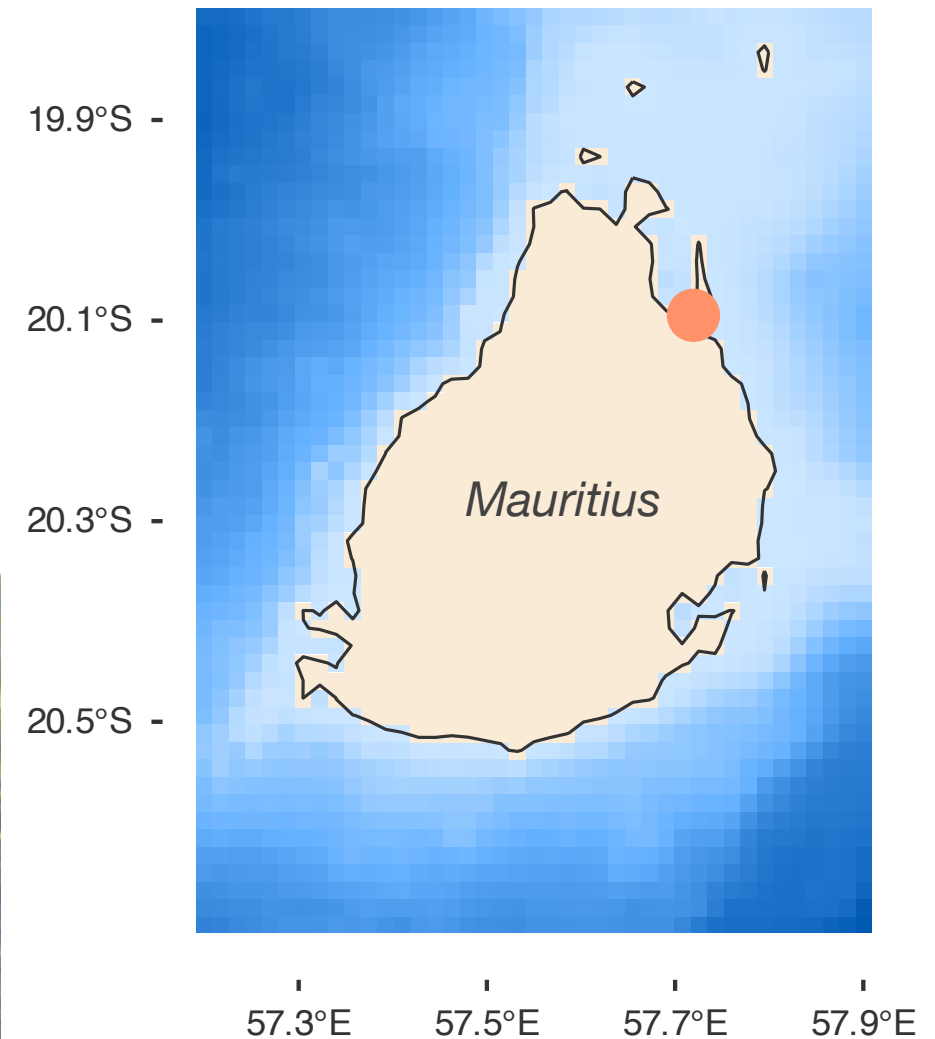


Image by Reef Conservation
Globe by Yashveer Poonit, CC BY-SA 3.0

Sampling scheme and sequencing

2 Sites

- Roches Noires
- Bel Ombre

3 Habitats

- Bare sand
- Seagrass meadow
- Channel to open ocean

2 Seasons

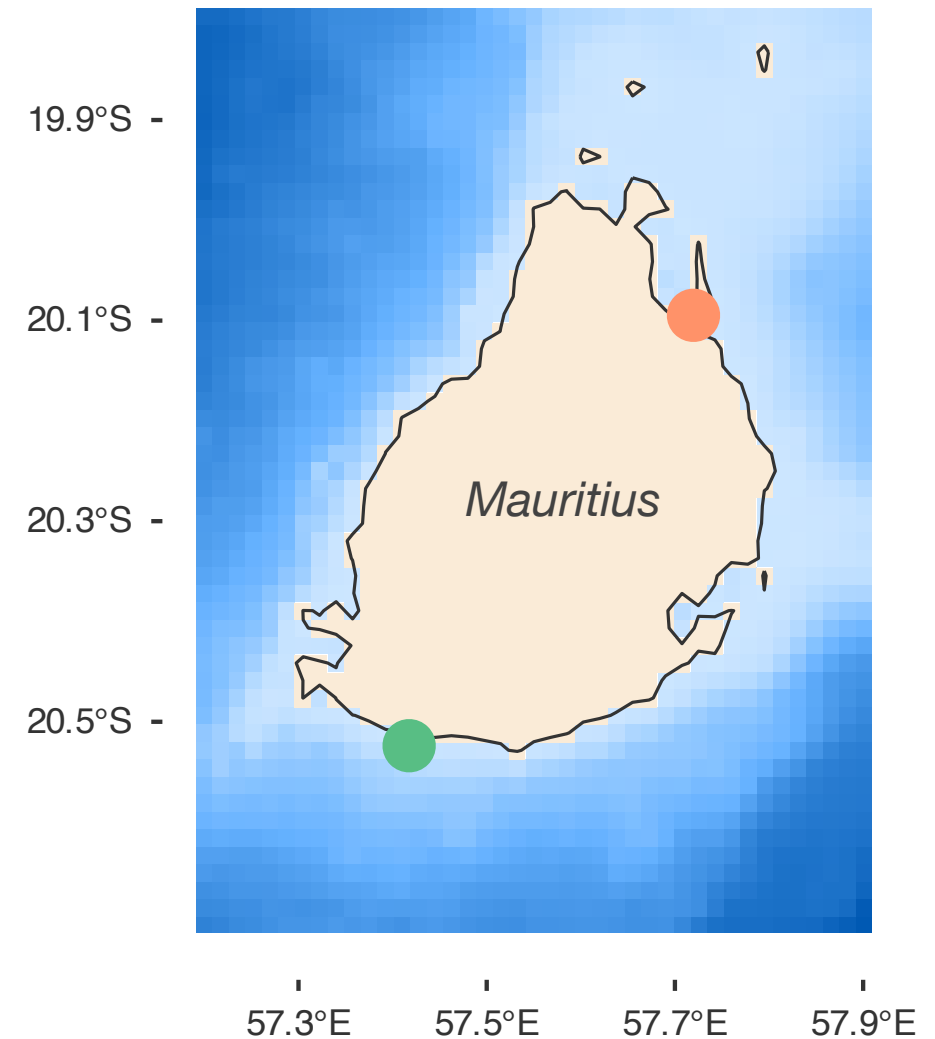
- summer
- winter

3 Replicates each

Method

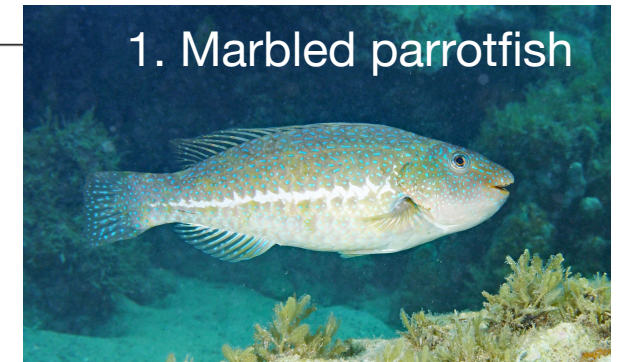
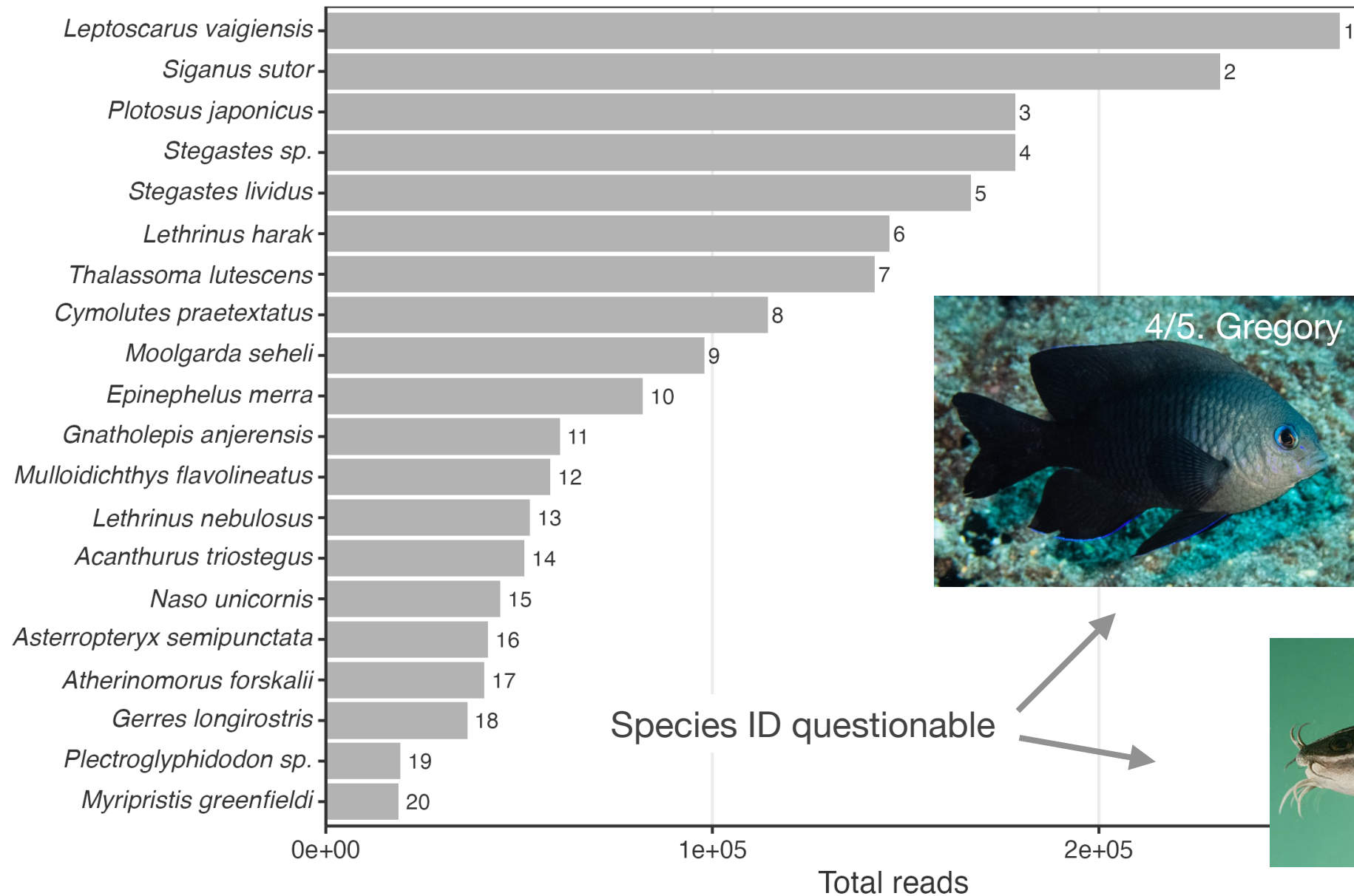
- Filtered sea water
- 12S (MiFish)
- Illumina MiSeq

Dataset



Most abundant species overall

Exercise 4



1. by RJ Koch | 2. by J Travis | 3. by JE Randall (all FishBase), 4/5. by Bill Ludt (NHM LA)

Diversity metrics

Exercise 1/2

How diverse is the community?

How similar / dissimilar are the communities?

Alpha diversity

Within-community diversity, e.g. species richness, **Shannon's index** — $H' = - \sum_{i=1}^R p_i \ln p_i$
(richness + evenness)

Beta diversity

Dissimilarity between communities, e.g. Jaccard distance (presence / absence only),

Bray–Curtis index (relative abundances)

$$\text{— } BC_{ij} = 1 - \frac{2C_{ij}}{S_i + S_j} \quad \begin{array}{l} 0 : \text{same community} \\ 1 : \text{no similarity} \end{array}$$

Principal coordinate analysis based on Bray–Curtis

Exercise 2

- Calculates **Bray–Curtis dissimilarities** between all pairs of samples based on taxon abundances (sample × sample matrix)
- Uses PCoA to reduce the dissimilarity matrix to a few **principal coordinates** (axes)
- Principal coordinate 1 explains the **most variation** in community dissimilarity, followed by PCo 2 etc.
- Plots samples as points in **two-dimensional space** based on their coordinates (samples are positioned so that 2D distances approximate their Bray–Curtis dissimilarities)
- ▶ Reveals **patterns in community composition**, such as clustering by habitat or site, environmental gradients, and outliers

Summary

[illegible]

Key takeaways

Summary

- Alpha diversity describes **how diverse** individual communities are across sites or conditions (within-sample biodiversity)
- Beta diversity reveals whether samples **differ in their species composition** and which factors may shape it (between-sample biodiversity)
- Alpha diversity requires standardizing read depth and retaining rare species, whereas beta diversity may be biased by rare species (adjust filtering approach)
- **Community composition complements diversity metrics** by showing which taxa drive the observed patterns and how their relative abundances vary across samples

Please help to [evaluate this course](#)

June 26 – July 17

